

Proposal for Orbital 25

Team Name:

EcoTrack

Proposed Level of Achievement:

Apollo 11

Motivation

Waste management is a pressing global issue, with millions of tons of waste being improperly disposed of each year. Recycling is a crucial part of sustainable waste management, but many people lack the motivation and awareness to participate actively. Our project aims to address this gap by incentivizing recycling through a reward system. By gamifying the recycling process and providing users with tangible benefits, we hope to encourage more responsible waste disposal and promote a circular economy.

Aim

We hope to develop an intuitive web application that allows users to track their recycling activities, earn points based on their contributions, and redeem rewards. The platform will also provide valuable insights into global recycling efforts, facilitate access to waste disposal locations, and engage users with an interactive information and feedback system.

User Stories

User Type	Objective	Task in System
Eco-conscious citizen	Contribute to sustainability	Track recycling efforts and earn points to stay motivated and be rewarded
Local government/waste management agency	Analyze recycling trends and participation	Access recycling data to develop a better waste management policy
Businesses interested in sustainability	Align brand with environmental responsibility	Offer incentives for recycling efforts to attract eco-friendly customers

Features

Core Features

User Account System

Users can create and manage their accounts. They will be prompted to log in using their email and password, with an option to reset passwords through email verification. Sign-ups require a username,

email, and password. Once logged in, users can update their profile, change credentials, or delete their account. Authentication and data storage will be handled by Firebase Authentication.

Recycling Activity Tracking and Reward System

Users earn points for logging recycling activities based on waste type and quantity. These points can be redeemed for discounts or prizes.

Waste Management Locations

An interactive map helps users locate nearby recycling centers and waste disposal points using Google Maps API.

Global Recycling Milestones

A leaderboard displays recycling efforts across countries, offering real-time insights into national and international waste management progress.

Extension Features

Reward System Expansion

Businesses can participate by offering eco-friendly incentives for users who accumulate points. Users can redeem these for vouchers and discounts.

Information & Feedback Center

A knowledge hub provides proper recycling guidelines. Users can submit feedback on waste management issues and report incorrect disposal practices.

Timeline

12-19 May - Lift-off (Programme Overview + Idea Formulation)

- Conduct initial brainstorming and refine project scope.
- Research existing waste management solutions and identify gaps.
- Finalize the feature list and technical requirements.

2 Jun - Evaluation Milestone 1 (Ideation)

- Complete project proposal and initial wireframes.
- Set up the development environment, including repository setup.
- Implement a basic user authentication system.

30 Jun - Evaluation Milestone 2 (Prototyping)

- Develop the core features: user account system, recycling activity tracking, and waste management location services.
- Implement the points system and ensure data is correctly stored in the database.
- Build and test UI components for seamless navigation.

28 Jul - Evaluation Milestone 3 (Extension)

- Integrate additional features like the reward redemption system.
- Develop the information and feedback center.
- Optimize UI/UX based on initial feedback.
- Conduct functional testing and bug fixes.

27 Aug - Splashdown (Refinement & Final Submission)

- Conduct final refinements and performance testing.
- Prepare the physical poster and documentation for presentation.
- Participate in the award ceremony and final showcase.

Tech Stack

Frontend: React.js – Handles the dynamic UI components and manages user interactions efficiently.

Backend: Node.js with Express – Powers API development, authentication, and data processing.

Database: PostgreSQL – A relational database for structured storage of user profiles, recycling records, and reward transactions.

ORM: Prisma – Simplifies database interaction and ensures efficient queries.

Authentication: Firebase Authentication – Ensures secure user logins and account management.

Cloud Storage: AWS S3 / Firebase Storage – Stores images and recycling proof submissions.

Geolocation Services: Google Maps API – Helps users locate nearby waste management facilities.

Hosting & Deployment: Vercel (frontend) and Railway / AWS RDS (backend + database) – Ensures stable deployment and scalability.

Testing: Jest / Mocha – Used for backend unit and integration testing.

Frontend Testing: Cypress – Automates UI testing to ensure reliability across different devices.

Design & Prototyping: Figma – Used to create wireframes and UI mockups.

Documentation & Architecture: Swagger for API documentation, Lucidchart / Draw.io for system diagrams.

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Qualifications

Our team consists of beginner to intermediate-level developers who are eager to apply and expand our technical knowledge through this project. While we may not have extensive industry experience, we have foundational skills in full-stack web development, cloud computing, and database management that will enable us to execute this project effectively.

Technical Skills

- **Frontend Development (React.js, HTML, CSS, JavaScript)**
We have experience building interactive web applications using React.js, implementing dynamic UI elements, and managing state efficiently.
- **Backend Development (Node.js, Express, PostgreSQL, Prisma)**
Our team is comfortable setting up RESTful APIs, managing authentication, and structuring relational databases using PostgreSQL with Prisma ORM for efficient data handling.
- **Version Control & Collaboration (Git, GitHub)**
We actively use Git for version control and follow a structured branching workflow to maintain code quality and allow seamless collaboration.
- **Cloud & Deployment (Vercel, Railway, AWS RDS, Firebase Storage)**
We have hands-on experience with deploying web applications using cloud platforms such as Vercel and Railway while managing backend infrastructure using AWS RDS and Firebase Storage.
- **Testing & Debugging (Jest, Mocha, Cypress)**
Our team follows test-driven development (TDD) principles to ensure application reliability, utilizing Jest/Mocha for backend testing and Cypress for frontend UI testing.
- **UI/UX Design (Figma, Wireframing, Prototyping)**
We use Figma to design wireframes, prototypes, and user interfaces to create a seamless and intuitive user experience.

Previous Project Experience

While this is our first major collaborative project, individual members of our team have worked on the following:

1. Vincent Gavriel Julijanto (A0305748X)

Technical Background

- **Full-Stack Web Development Bootcamp** – Covered a range of technologies including HTML, CSS, JavaScript, Node.js, React, PostgreSQL, Web3, and DApps, providing hands-on experience in building full-stack applications.
- **CS50: Introduction to Computer Science** – Gained foundational knowledge in programming, algorithms, and data structures.
- **React Course** – Developed hands-on expertise in building scalable and interactive user interfaces using React.js.

Project & Hackathon Experience

- **Third Prize, NUS Fintech Hackathon** – Developed a cryptocurrency fraud detection system for an investment bank in Singapore. The system leveraged data analytics and machine learning to identify suspicious transactions and mitigate fraudulent activities in cryptocurrency trading.

- **Ongoing Health Hack Hackathon** – Currently working on a healthcare-related solution, gaining experience in building applications that optimize processes and improve user experiences.

2. Steven Nathanael Mulyadi (A0300549J)

Technical Background

- **CS1010A : Introductory Course to Python Programming**
- **CS2030 : Java Programming course**
- **Learnt about Blockchain System individually**

Learning and Adaptability

Since we are beginner to intermediate-level developers, we acknowledge that learning will be a key part of our journey. To bridge any knowledge gaps:

- We will follow structured online courses and documentation on **React.js, Node.js, PostgreSQL, and Prisma.**
- We will seek guidance from mentors and experienced developers in the Orbital 25 program.
- We will conduct frequent team discussions and debugging sessions to troubleshoot technical challenges collaboratively.

Our team is committed to building a high-quality, functional web application that aligns with the Artemis standards while continuously improving our technical competencies.

Software Engineering

To ensure a structured and efficient development process, we plan to adopt the following software engineering practices:

1. Agile Development

We will follow an agile methodology, implementing bi-weekly sprints for iterative development and rapid feedback integration. This ensures that the team can continuously refine features and respond to challenges effectively.

2. Version Control with Git & GitHub

All code will be managed using Git with GitHub as our central repository. We will follow a branching strategy (feature branches, development, and main) to maintain code quality, and all changes will be reviewed through pull requests before being merged.

3. Code Reviews

To maintain high code quality, regular peer reviews will be conducted. Team members will provide constructive feedback on logic, efficiency, and best practices before code is integrated into the main branch.

4. Unit & Integration Testing

- **Backend:** Jest / Mocha will be used to write unit tests for API endpoints and core business logic.
- **Frontend:** Cypress will be implemented to automate UI interaction tests and ensure user workflows function correctly.
- **Integration Tests:** End-to-end tests will be conducted to verify communication between frontend and backend services.

5. Continuous Integration & Deployment (CI/CD)

Automated pipelines will be set up for both frontend and backend using GitHub Actions.

This will allow us to deploy updates seamlessly to Vercel (frontend) and Railway / AWS RDS (backend + database) whenever code changes are approved and merged.

6. Documentation

- **API Documentation:** Swagger will be used to document all API endpoints clearly so that developers can understand how to interact with them.
- **Technical Documentation:** A Confluence or Notion workspace will be maintained with system architecture details, database schema, and project decisions.

7. Security Best Practices

- **Authentication & Authorization:** Firebase Authentication will be implemented to manage secure user logins, including OAuth options for social sign-ins.
- **Data Protection:** Sensitive user data will be encrypted, and access control policies will be set for different API endpoints to prevent unauthorized data exposure.
- **Regular Security Audits:** The team will conduct code scans and dependency checks using tools like Snyk or Dependabot to identify vulnerabilities.

8. Scalability & Performance Optimization

- The backend will be structured with modular services to allow for horizontal scaling when needed.
- Database indexing and caching strategies will be applied to optimize response times for frequently accessed queries.
- Static assets and frontend content will be optimized using modern bundling techniques to ensure quick page loads.

By adhering to these engineering practices, our team will ensure that the project is developed efficiently, remains maintainable, and meets industry best practices despite our beginner-level experience.